

Meghal Dani

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RESEARCH INTEREST

Computer Vision, Machine Learning, Deep Learning, Foundation Models, Large Language Models, Safe AI in healthcare, Computational Neuroscience, Multi-modal learning (text, biosignals, medical images, videos), Explainable AI

EDUCATION

Ph.D. in Computer Science <i>International Max Planck Research School for Intelligent Systems, University of Tuebingen</i>	Aug. 2022 – Present
Masters of Technology in Computational Biology <i>Indraprastha Institute of Information Technology Delhi (IIIT-D)</i>	Jul. 2017 – Jul. 2019 GPA: 9.55/10 (<i>Distinction</i>)
Bachelor of Engineering in Computer Science <i>Birla Institute of Technology (BIT, Mesra)</i>	Jul. 2012 – May. 2016 GPA: 8.47/10 (<i>Distinction</i>)

SELECTED RESEARCH EXPERIENCE

Visiting Researcher, Neuroscience Institute, Groote Schuur Hospital <i>Cape Town, South Africa</i> • <u>Advisors</u>: Dr. MV Gule and Dr. rer. nat. Stefanie Liebe • Developing AI pipelines to extract seizure types and onset zones from handwritten epilepsy records, while uncovering cross-cultural variations in symptoms and clinical workflows across South African cohorts.	Nov. 2025 – Dec. 2025
Researcher, TCS Research and Innovation Labs <i>New Delhi, India</i> • <u>Advisors</u>: Dr. Lovekesh Vig, Ramya Hebbalaguppe • Developed anchor-free universal lesion detection network for CT scans achieving 86.05% sensitivity, published at BMVC and ISBI with a patent • Created lightweight 3D pose estimation using skeletal graphs with $11\times$ space and $3\times$ time reduction, published at SIGGRAPH and WACV with a patent	Aug. 2019 – Aug. 2021
Graduate Student Researcher, Image Analysis and Biometrics Lab (IAB) <i>New Delhi, India</i> • <u>Advisors</u>: Prof. Dr. Richa Singh and Prof. Dr. Mayank Vatsa • Masters thesis nominated for the best thesis award	Aug. 2018 – Jul. 2019
Research Intern – TCS Research and Innovation Labs <i>New Delhi, India</i> • <u>Advisors</u>: Ramya Hebbalaguppe • This work got accepted at ISMAR.	May 2018 – Jul. 2018

PUBLICATIONS, ABSTRACTS, AND PATENTS

- **M. Dani**, M. J. Prakash, Z. Akata, and S. Liebe, “*SemioLLM: Evaluating Large Language Models for Diagnostic Reasoning from Unstructured Clinical Narratives in Epilepsy*,” **Nature Communications Medicine** 2025 (Also presented at **ICML AI for Science workshop** 2024)
- **M. Dani**, M. J. Prakash, Z. Akata, and S. Liebe, “*Towards Automatized Analysis of Epileptic Seizure Behavior in VEEG Recordings*,” **Bernstein Conference-Abstract** 2023
- **M. Dani***, I. Rio-Torto*, S. Alaniz, and Z. Akata, “*Devil: Decoding vision features into language*,” in **DAGM GCPR (Oral)**, Springer, 2023, pp. 363–377. (Also presented at **ICCV-CLVL workshop** 2023)
- M. Sheoran*, **M. Dani***, M. Sharma, and L. Vig, “*An efficient anchor-free universal lesion detection in ct-scans*,” in 2022 IEEE 19th **International Symposium on Biomedical Imaging (ISBI)**, IEEE, 2022, pp. 1–4.
- M. Sheoran*, **M. Dani***, M. Sharma, and L. Vig, “*DKMA-ULD: Domain knowledge augmented multi-head attention based robust universal lesion detection*,” **BMVC**, pp. 363–377, 2022.
- **M. Dani**, K. Narain, and R. Hebbalaguppe, “*3DPoseLite: A compact 3d pose estimation using node embeddings*,” in **WACV**, Springer, 2021, pp. 1878–1887

- **M. Dani**, A. Popli, and R. Hebballaguppe, “*PoseFromGraph: Compact 3-d pose estimation using graphs*,” in **SIGGRAPH**, 2020, pp. 1–4.

Patents

- U.S. Patent 12131466: M. Sheoran*, **M. Dani***, M. Sharma, and L. Vig, “*Method and system for domain knowledge augmented multi-head attention based robust universal lesion detection*,” , Oct. 2024
- U.S. Patent 12033352: R. Hebballaguppe, **M. Dani**, and A. Popli, “*Methods and systems for generating end-to-end model to estimate 3-dimensional (3-D) pose of object*,” , Jul. 2024

RELEVANT PROJECTS

EEG foundation models	2025
<i>Technologies Used:</i> <i>EEG foundation models pretraining, self-supervised learning (MAE), Representation Learning</i>	
• Investigating how pretraining strategies, data scale, augmentation of signals affect the performance of EEG foundation models for BCI tasks.	
Brain Research International Data Governance & Exchange (BRIDGE)	2025
<i>Technologies Used:</i> <i>MRI data preprocessing, Neuroimaging QC, Statistical analysis</i>	
• Supported development of international data governance frameworks by systematically accessing, preprocessing, and analyzing multi-country brain MRI datasets (SUDMEX_CONN and Nigerian Brains) while documenting legal, technical, and training barriers to cross-border data sharing.	
Seizure Detection Challenge (SzCORE, EPFL)	2025
<i>Technologies Used:</i> <i>PyTorch, State-Space Models (S4, Hyena), EEG Segmentation</i>	
• Developed S4 and Hyena-based state-space model architectures for EEG-based seizure detection and segmentation, achieving 3rd position out of 30 submissions in the SzCORE epilepsy benchmark challenge.	
Towards Automatized Analysis of Epileptic Seizure Behavior in VEEG Recordings	2024
<i>Technologies Used:</i> <i>PyTorch, OpenCV, FFmpeg, MediaPy, Computer Vision</i>	
• Developed a pose-estimation framework using TAPIR that tracks 21 body joints from a single camera view in low-light, low-resolution hospital videos with heavy occlusions (blankets, staff, equipment), generating joint pose trajectories to detect seizure vs non-seizure events and classify seizure type.	

CONTRIBUTIONS TO RESEARCH GRANTS

- Contributor (Discussions, Review, Editing) BMBF : DeepSync (2025), Dr. rer. nat. Stefanie Liebe, Prof. Dr. Andreas Geiger, Prof. Dr. Jakob Macke, Prof. Dr. Max Welling - Accepted
- Co-Lead Contributor [DFG Initiative for a 2-stage Initiation of International Collaboration \(I2I\)](#)(2024), Dr. rer. nat. Stefanie Liebe, Dr. Manqoba Vusumuzi Gule (University of Cape Town, South Africa) - Accepted

TECHNICAL SKILLS

Programming Languages: Python, R, MATLAB, SQL, C/C++, Java, HTML, CSS

ML/DL/CV Frameworks: PyTorch, OpenCV, HuggingFace, LangChain, scikit-learn, FFmpeg, Pandas

Tools and Libraries: Linux, Git, Bash, Inkscape, SLURM, HPC Servers, L^AT_EX, Jupyter, Matplotlib

PAST ACHIEVEMENTS, AWARDS AND COMMUNITY SERVICE

- Secured third position on **epilepsy benchmark challenge** to detect seizures from EEG data held by EPFL among 30 participating teams (2025)
- [Soft Skills Seminar Series \(S4\) Workshop](#) Organizer at IMPRS-IS (2024 - Present)
- **Ph.D. Representative** in Search Committee for Tenure-Track Professor of Machine Learning and Intelligent Systems, University of Tuebingen (2024)
- **Awarded ClinBrAI n PhD Fellowship:** by EKFS, Tuebingen, Germany (2022)
- **Best Thesis Award Nominee:** for masters thesis at IIIT-Delhi (2019)
- **Travel Award** from IIIT-D to present at International Symposium on Mixed and Augmented Reality (ISMAR) (2018)
- **Postgraduate Fellowship:** Received funding from the Government of India (2017)
- **Academic Reviewer:** ICCV 2023 - 2025, CVPR 2025, ICML 2024, MICCAI 2024, ACCV 2024